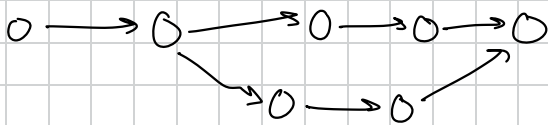


Error - Prone Reads



Bubbles! Happens because of read breaking.
This can be caused by inexact repeats.
Repeated region in genome with minor variations.

Genome is 23 pairs. Heterozygosity (maybe has slight nucleotide difference)

Somatic mutations (accumulate as you age).

The path in a bubble produced from an error is likely to be short and not common.

Direction of DNA strand is opposite

The two networks are amalgamated (reverse complement and normal)

Finding Hidden Messages in DNA

Replication begins in a region called the replication origin (denotes ori)

How do we find this?

Replication is initiated by DnaA, binds to a short segment known as a ^{DNA} box

Multiple DnaA boxes → higher chance of binding → higher "fitness"

When looking for boxes, find frequent words in all windows within a genome.

Problem: Given string genome size, window length l , k -mer length, threshold

A k -mer forms an (L, f) -clump inside Genome if it appears at least f times in a L -length window of Genome

However, genomes have many repeats.

E-Coli has over 1900 different 9-mers from (500, 3)-clumps

DNA polymerase goes from 3' to 5'

In single-stranded DNA, C rapidly mutates into T through deamination
While lagging, the C might change to a T